

# Local Floer Chain Complex across Hamiltonian Bifurcation

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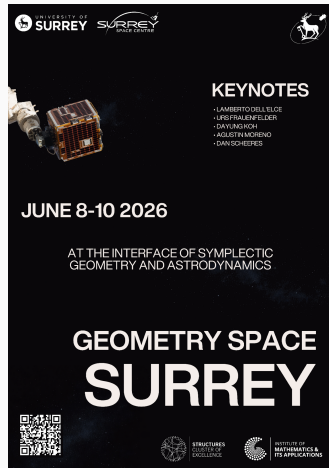


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# Recurrence?



(a) Childhood dream...



(b) came true, in some sense.

This talk is based on ongoing joint work with Hong-kwon Jo (SNU).

**Main result.** We describe the explicit changes in Conley–Zehnder indices and the Floer chain complex for generic types of Hamiltonian bifurcations.

- **Motivation and Preliminaries.**

Bifurcations, the Conley–Zehnder index, and Floer homology.

- **Classification of Hamiltonian bifurcations.**

Birth-death, period doubling, phantom kiss and emission.

- **Chain complex mutation.**

Construction of cylinders, Morse–Bott cascade counting.

# Motivation

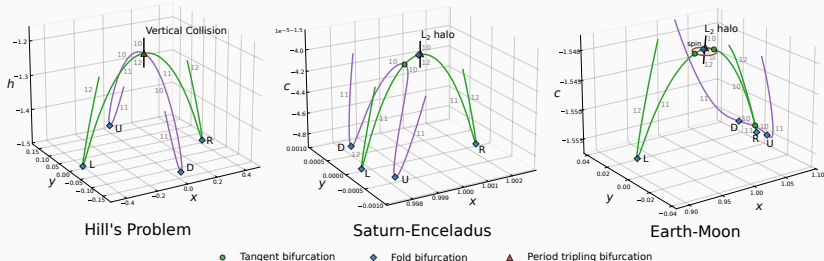
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# Motivation 1. Bifurcations

A **bifurcation** is, heuristically, the birth or death of periodic orbits.

Several recent works [AFvK<sup>+</sup>23], [AB25], [JKvK26] relate bifurcations to symplectic invariants, for example Conley–Zehnder indices, Floer-type Euler characteristics, etc.



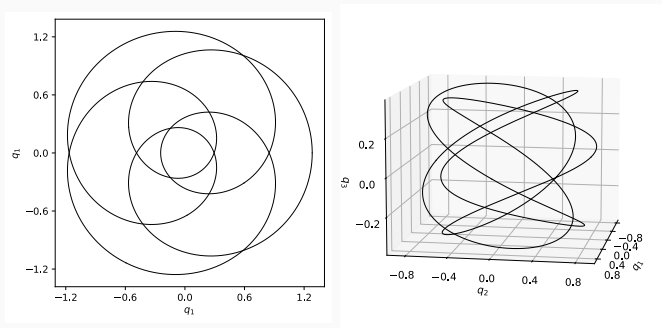
**Figure 2:** Bifurcation in Hill's lunar problem and CRTBP, from [JKvK26].

Meyer's classical work [Mey70] classified generic bifurcation types. Can we make a *symplectic dictionary* for these bifurcation types?

## Motivation 2. Rotating Kepler Problem

In the rotating Kepler problem, there are three important periodic orbits: retrograde, direct and vertical collision orbits.

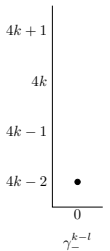
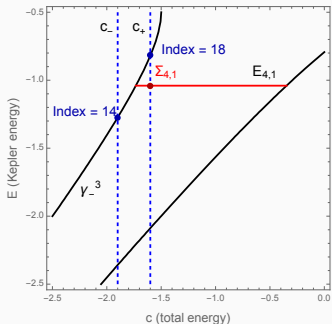
The multiple covers of the retrograde and direct orbits bifurcate, producing orbits that look like flowers.



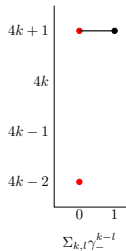
**Figure 3:** Resonant orbits of RKP, courtesy of Chankyu Joung.

## Motivation 2. Rotating Kepler Problem

In my previous work [Lee26], I observed that this bifurcation can be translated into a change in the Floer chain complex.



At  $c_{k,l}^- - \epsilon$



At  $c_{k,l}^- + \epsilon$

**Slogan.** A bifurcation gives a *mutation of a Floer chain complex*.

We aim to describe this effect explicitly.

# Preliminaries

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# Bifurcation

Let  $H_\varepsilon : W^4 \rightarrow \mathbb{R}$  be a 1-parameter family of Hamiltonians, and let  $\gamma_0$  be a periodic orbit of  $H_0$  with Floquet multiplier  $\lambda_0$ , an eigenvalue of the linearized return map on the Poincaré section.

If  $\lambda_0 \neq 1$ , so the orbit is **non-degenerate**, then nearby periodic orbits form an *orbit cylinder*

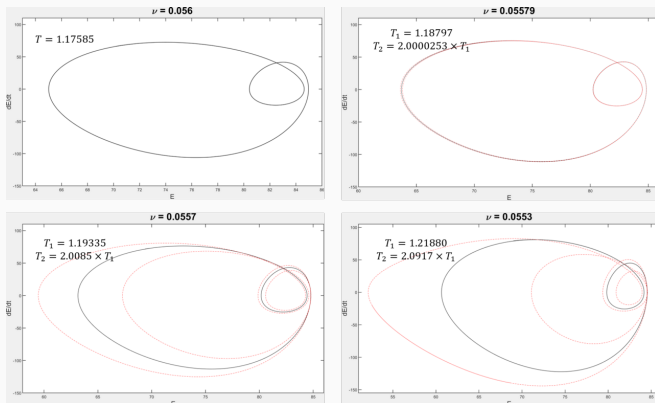
$$\Gamma = \{(\gamma_\varepsilon, \varepsilon) : \gamma_\varepsilon \text{ is a periodic orbit of } H_\varepsilon\}.$$

This is a *continuation* of the orbit along the bifurcation parameter.

We follow  $\gamma_\varepsilon$  and detect  $\lambda_\varepsilon$ . If  $\lambda_\varepsilon = 1$  at some  $\varepsilon$ , then  $\gamma_\varepsilon$  is degenerate, and the orbit cylinder may break:

- periodic orbits may be born or vanish; this is called a **bifurcation**,
- if  $\lambda \neq 1$  but  $\lambda^k = 1$ , then the  $k$ -th cover bifurcates.

## Example. Period Doubling



**Figure 4:** Period doubling in Kuramoto–Sivashinsky equation, from [Max21].

Period doubling is a generic bifurcation of a double cover.

The red dashed orbit is the new orbit that emerges from the bifurcation.

# Conley–Zehnder Index

The **Conley–Zehnder index**  $\mu_{CZ}(\gamma)$  is an integer associated with a periodic orbit and can be understood as a *symplectic winding number*.

We are interested in the index along the parameter;  $\mu_{CZ}(\gamma_\epsilon)$ .

Along a non-degenerate family, the CZ index is constant.

**Key idea.** A change in  $\mu_{CZ} = \text{Degeneracy of } \gamma_\epsilon = \text{A bifurcation occurs}$

Hence, a bifurcation can be detected via change of the CZ index.

# Floer Homology

**Floer homology**  $HF(W, H)$  is a topological invariant associated with a symplectic manifold  $W$  and Hamiltonian  $H$ . (cf. [AD14].)

This is determined by the **Floer chain complex**  $CF(W, H)$ , whose

- generators are 1-periodic orbits of  $H$ ,
- grading is given by CZ indices,
- the differential is determined by counting **Floer cylinder**, whose boundaries are asymptotic to periodic orbits.

Across a bifurcation,  $CF(W, H)$  changes substantially;

- orbits are born or vanish  $\Rightarrow$  the number of generators changes.
- the CZ index changes  $\Rightarrow$  grading changes

However, the homology doesn't change, so the differential is responsible for keeping the homology invariant. *Can we describe this explicitly?*



# Classification of Bifurcations

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# Meyer's Classification

Generic Hamiltonian bifurcations are classified in [Mey70].

They are organized by the covering number  $k$  of the underlying orbit.

- $k = 1$ : birth-death
- $k = 2$ : period doubling, materialization (opposite stability types)
- $k = 3, 4$ : phantom kiss
- $k \geq 4$ : emission

**Aside.** We will use names from [AM78].

Three extra types are given, which include critical points.

Other names: touch-and-go, period tripling/quadrupling, etc.

# Birkhoff Normal Form

We find the periodic points of the iterated return map  $\Phi_\varepsilon^k$ .

This can also be done in essentially the same way, by finding critical points of the generating Hamiltonian of the return maps.

1. We can write the generating Hamiltonian of the return map  $\Phi$  by

$$G_t = G_2 + \sum_{k \geq 3} G_{k,t}$$

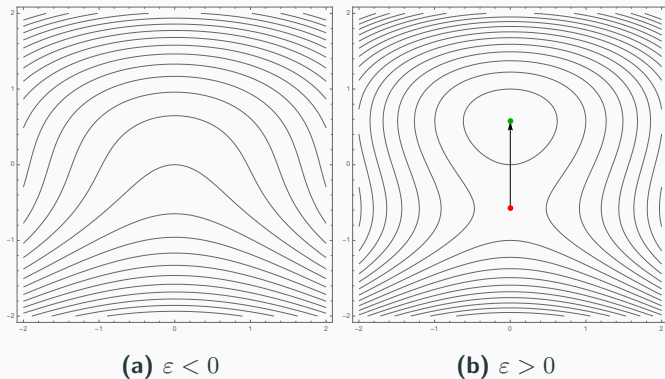
where  $\deg G_k = k$ ,  $G_2^{ss} = \alpha(u^2 + v^2)$ ,  $\dot{G}_{k,t} = \{G_2^{ss}, G_{k,t}\}$ ,  $\lambda = e^{i\alpha}$ .

2. If  $\lambda_0^k = 1$ , the generating Hamiltonian of  $k$ -th iterate of the return map  $\Phi^k$  can be chosen with  $G_2^{ss} = 0$ , so  $\dot{G}_{k,t} = 0$ .
3. We want  $G/k$  to generate  $\Phi$ , which forces a symmetry of  $G$ .

# Generating Hamiltonians – Birth-death

A generic single cover bifurcation, with generating Hamiltonian

$$G_\varepsilon(u, v) = \alpha v^2 + \beta u^3 + \dots + \varepsilon(u + \dots).$$

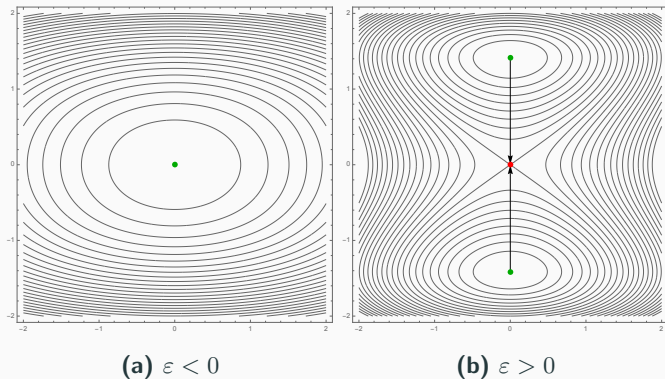


**Figure 5:** Critical points and gradient trajectories of birth-death bifurcation

# Generating Hamiltonians – Period doubling

A generic double cover bifurcation, with generating Hamiltonian

$$G_\varepsilon(u, v) = \alpha v^2 + \beta u^4 + \dots + \varepsilon(u^2 + \dots).$$

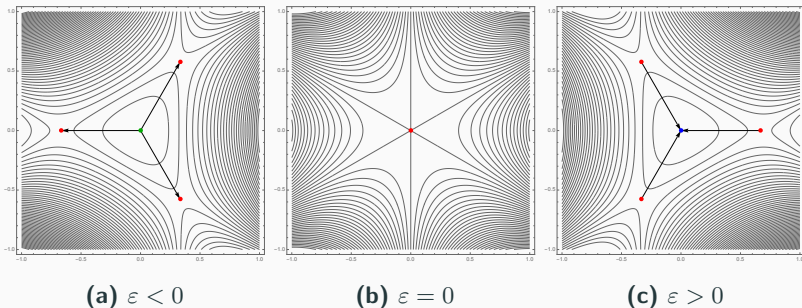


**Figure 6:** Period doubling bifurcation

# Generating Hamiltonians – Phantom kiss

A generic 3rd or 4th cover bifurcation, with generating Hamiltonian

$$G_\varepsilon(r, \theta) = \alpha r^k \cos k\theta + \dots + \varepsilon(r^2 + \dots).$$

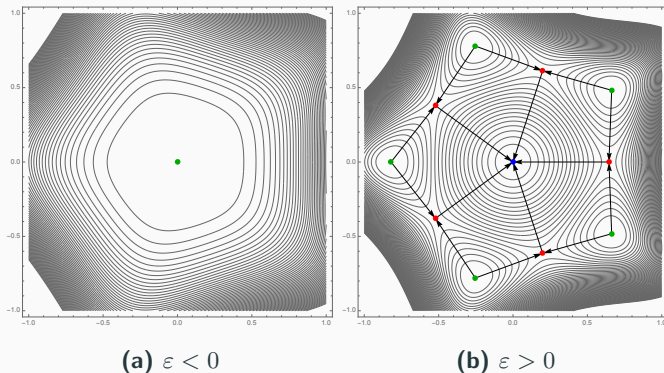


**Figure 7:** Phantom kiss bifurcation

# Generating Hamiltonians – Emission

Generic  $k \geq 4$ -th cover bifurcation, with generating Hamiltonian

$$G_\varepsilon(r, \theta) = \alpha r^4 + \beta r^k \cos k\theta + \dots + \varepsilon(r^2 + \dots).$$



**Figure 8:** Emission bifurcation

**Note.** The fourth cover is a borderline case, depending on  $|\alpha|$  and  $|\beta|$ .

# Summary of the Classification

| Name            | Floquet Mult.                 | Before     | Stability  | Morse Index | After      | Stability  | Morse Index |
|-----------------|-------------------------------|------------|------------|-------------|------------|------------|-------------|
| Birth-death     | $\lambda = 1$                 |            |            |             | $\gamma_E$ | Elliptic   | 0 or 2      |
|                 |                               |            |            |             | $\gamma_H$ | Hyperbolic | 1           |
| Period doubling | $\lambda = -1$                | $\gamma^2$ | Elliptic   | 0 or 2      | $\gamma^2$ | Hyperbolic | 1           |
|                 |                               |            |            |             | $\gamma_E$ | Elliptic   | 0 or 2      |
| Materialization | $\lambda = -1$                | $\gamma^2$ | Hyperbolic | 1           | $\gamma^2$ | Elliptic   | 0 or 2      |
|                 |                               |            |            |             | $\gamma_H$ | Hyperbolic | 1           |
| Phantom kiss    | $\lambda^k = 1$<br>$k = 3, 4$ | $\gamma^k$ | Elliptic   | 0 or 2      | $\gamma^k$ | Elliptic   | 2 or 0      |
|                 |                               | $\gamma_H$ | Hyperbolic | 1           | $\gamma_H$ | Hyperbolic | 1           |
| Emission        | $\lambda^k = 1$<br>$k \geq 4$ | $\gamma^k$ | Elliptic   | 0 or 2      | $\gamma^k$ | Elliptic   | 2 or 0      |
|                 |                               |            |            |             | $\gamma_H$ | Hyperbolic | 1           |
|                 |                               |            |            |             | $\gamma_E$ | Elliptic   | 0 or 2      |

**Table 1:** Generic Hamiltonian bifurcations

A change of Morse indices is directly related to a change of CZ indices.

**Note.** There are more bifurcations, (e.g. pitchfork), appear in the symmetric system, but if the symmetry is discrete, (e.g. CRTBP), the classification is essentially based on these basic types.



## Chain Complex Mutation

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## An easy example. Birth-death bifurcation

Suppose an orbit  $\gamma_0$  experiences the birth-death bifurcation along  $\varepsilon$ .

1. If  $\varepsilon < 0$ , there are no orbits.
2. If  $\varepsilon > 0$ , there are two orbits, say  $\gamma_H$  and  $\gamma_E$ .

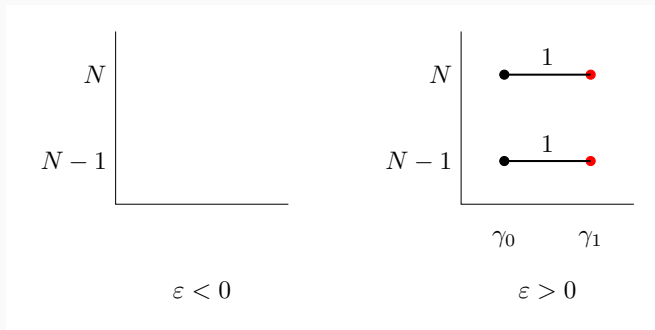
### Floer-theoretic viewpoint.

1. If  $\varepsilon < 0$ ,  $CF = 0$ .
2. If  $\varepsilon > 0$ ,  $CF = \langle \gamma_H, \gamma_E \rangle$ .
3. Both cases should produce the same (local) Floer homology.

Hence,  $|\mu_{CZ}(\gamma_H) - \mu_{CZ}(\gamma_E)| = 1$  and they should cancel each other out, so there exists a Floer cylinder between them. (cf. [GG10].)

However, the situation is more complicated for higher covers.

## An easy example. Birth-death bifurcation



**Figure 9:** Chain complex mutation across the birth-death.

- Points at  $(n, m)$  corresponds to a generator of degree  $n + m$ .
- The line indicates the differential.
- If the number is odd, it means two generators cancel each other.  
(We are using  $\mathbb{Z}_2$ -coefficients.)

# Main Result and Key Ideas

**Main result.** A full description of the mutation of Floer chain complex across the five generic bifurcation types.

## Key Ideas.

1. Construct a Hamiltonian  $H_\varepsilon$  represents a given bifurcation type.
2. Gradient trajectories of  $G_\varepsilon$  give Floer cylinders of  $H_\varepsilon$ .
3. Morse indices of critical points of  $G_\varepsilon$  give CZ index of orbits.  
( $\mu_{CZ}(\gamma_p) = N + (2 - \text{Ind}(p))$ )
4. We get cylinders connecting a simple orbit and a multiple cover.  
 $\Rightarrow$  A contribution of 1 cylinder may not be exactly 1.  
Using Morse–Bott cascades, we can compute the explicit chain complex and verify that its homology is invariant.

# Adiabatic approximation of Hamiltonian

Let  $\tilde{H}_\varepsilon$  be a Hamiltonian and  $G_\varepsilon$  be a generating Hamiltonian for the  $k$ -th iterate of the return map.

Using **adiabatic approximation**, which is an integration along the orbit, we construct the model Hamiltonian for each bifurcation type,

$$H_\varepsilon = h(r) + G_\varepsilon.$$

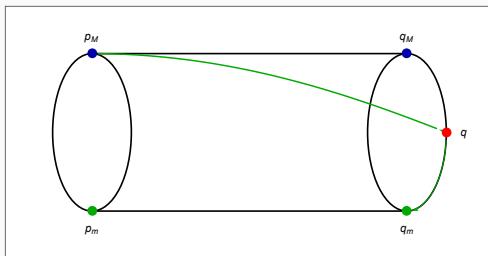
- $H_\varepsilon$  is defined on the covering space, but Floer-type information can descend to the base space.
- The rotating number, or CZ index, is encoded in the covering map.
- $H_\varepsilon$  is not exactly the same as  $\tilde{H}_\varepsilon$ , but has the same return map and Conley–Zehnder indices as  $\tilde{H}_\varepsilon$ .

# Morse–Bott Cascade

Our periodic orbits have an  $S^1$ -symmetry under time shifts, which makes the situation slightly different from the usual Floer theory.

We equip each orbit with a Morse function, and consider two critical points (maximum and minimum) as generators of a Floer chain complex.

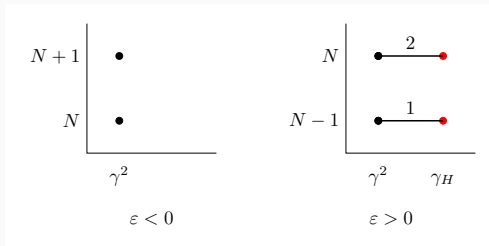
In this case, the differential is given by counting **Morse–Bott cascades**; a sequence of Morse trajectories and Floer cylinders.



**Figure 10:** Morse–Bott cascade.

# Chain complex mutation for Period doubling

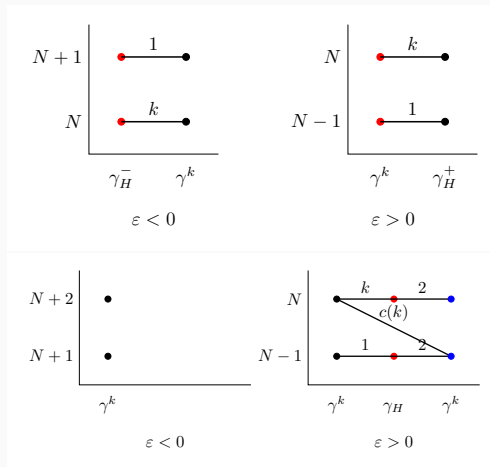
1. If  $\varepsilon < 0$ , there is one orbit  $\gamma^2$ .
2. If  $\varepsilon > 0$ , there are two orbits;  $\gamma^2$  and  $\gamma_H$ .
3.  $\mu_{CZ}(\gamma^2)$  changes, say, from  $N$  to  $N - 1$ .
4.  $\mu_{CZ}(\gamma_H)$  must be  $N$ , and there exists 1 Floer cylinder between  $\gamma^2$  and  $\gamma_H$ , whose contribution is as follows.



**Figure 11:** Chain complex mutation across the period doubling.

$\gamma_{H,\min}$  and  $\gamma_{\min}^2$  cancel each other, while  $\gamma_{H,\max}$  and  $\gamma_{\max}^2$  do not.

# Chain complex mutations for Phantom kiss and Emission



**Figure 12:** Chain complex mutations across phantom kiss and emission





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Illustration of period doubling in the Kuramoto–Sivashinsky equation. CC0 1.0 Universal Public Domain Dedication.



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**Figure 13:** Thank you for your attention!

Questions, comments, and further discussion are welcome!